

FretGarden

Pedagogical Constitution

Chapter 16

Scales, Chords, Harmony, and Musical Application

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This chapter defines how FretGarden turns theoretical structures into heard, playable, explainable, and creatively usable music.

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1. Purpose

This chapter governs pedagogy for scales, modes, chord construction, voicing, harmonic function, progression, voice leading, analysis, improvisation, composition, and repertoire application.

It ensures that larger musical structures are not reduced to diagrams, formulas, or isolated terminology.

2. Constitutional Authority

This chapter specializes all prior standards for integrated theory and musical application and is binding on scale, chord, harmony, improvisation, composition, and repertoire content.

3. Governing Principle

Governing Rule: Scales, chords, and harmony MUST be taught as coordinated systems of sound, interval structure, pitch content, fretboard realization, temporal behavior, function, and musical use.

4. Core Distinctions

A scale is not a fingering. A chord symbol is not a voicing. A voicing is not a function. Function is not destiny. Analysis is not application. Correct notes are not automatically musical phrasing.

5. Structural Domain Model

The domain includes collections, interval formulas, tonal centers, modes, chord formulas, inversions, voicings, extensions, progressions, cadences, function, voice leading, non-chord tones, modulation, and stylistic usage.

6. Scale Foundations

Scale instruction should coordinate pitch set, interval pattern, tonal center, degree function, sound, notation, and multiple fretboard realizations.

7. Scale Fingering Versus Scale Knowledge

Fingerings are performance solutions. Learners must identify roots, degrees, intervals, and connections within them and must transpose or alter them intentionally.

8. One-Octave and Multi-Octave Work

One-octave structures clarify interval order and degree identity. Multi-octave patterns develop range and continuity but must not obscure repeated scale structure.

9. Position and String-Set Variants

Scales should be played on one string, limited string sets, positions, connected positions, and along the neck as readiness permits.

10. Modes

Modes must be taught as collections with tonal centers, characteristic degrees, sound, harmony, and use. Rotating a major-scale pattern without establishing a new center is insufficient.

11. Pentatonic and Blues Collections

Common guitar collections should be connected to interval content, chord relationships, phrasing, stylistic vocabulary, and note targeting rather than presented only as boxes.

12. Chord Construction

Chords should be built from intervals and scale relationships. Formula memorization should be reinforced through spelling, hearing, playing, and alteration.

13. Triads

Major, minor, diminished, and augmented triads should be taught in root position and inversions across string sets, with note names, intervals, and voice-leading applications.

14. Seventh Chords and Extensions

Seventh chords and extensions should be introduced through audible function and interval structure. Omitted and doubled tones must be explained as voicing decisions.

15. Chord Symbols

Symbols encode root, quality, extension, alteration, bass, and sometimes voicing implications. Learners must distinguish what a symbol specifies from what remains open.

16. Voicing

Voicings are concrete pitch arrangements. Instruction should compare register, spacing, doubling, inversion, playability, and sound without treating one guitar grip as the chord itself.

17. Voice Leading

Voice leading should be taught through small pitch motion, common tones, guide tones, contrary motion, and melodic continuity between chords.

18. Harmonic Function

Tonic, predominant, dominant, and related functional models should be presented as analytical frameworks with stylistic limits, not universal laws.

19. Progressions and Cadences

Progressions should be heard, played, analyzed, transposed, and varied. Roman numerals and Nashville-style numbers must remain connected to actual keys and sounds.

20. Non-Chord Tones

Passing, neighbor, suspension, anticipation, and other tones should be understood through temporal and voice-leading behavior rather than memorized labels alone.

21. Borrowing, Secondary Function, and Chromaticism

Chromatic harmony should be introduced only when diatonic reference frames are sufficiently stable. Each device must be connected to audible effect and voice leading.

22. Analysis Standards

Analysis must answer a musical question. Labeling symbols without explaining relationships, expectation, structure, or performance implication is incomplete.

23. Improvisation

Improvisation tasks should combine constraints and agency: target chord tones, develop motifs, control rhythm, vary contour, create tension and release, and listen to accompaniment.

24. Composition

Composition tasks should require planning, revision, explanation, and auditory evaluation while permitting multiple valid outcomes.

25. Repertoire

Repertoire should function as evidence, context, and motivation. Examples must be legally and ethically usable and should represent varied styles without implying one canon defines musicianship.

26. Integration With Fretboard Knowledge

Every scale, chord, and harmonic task should reveal note, interval, degree, or chord-tone content at an appropriate stage. Fretboard navigation must serve musical decisions.

27. Integration With Rhythm

Harmonic knowledge must operate in time through chord changes, rhythmic comping, phrase placement, and form. Static shape accuracy is insufficient for performance claims.

28. Integration With Ear Training

Learners should hear collection color, chord quality, bass motion, cadence, and voice leading, then connect perception to names and actions.

29. Sequencing

Curriculum should spiral scales, chords, harmony, ear, rhythm, and application. Complete theoretical coverage is not required before meaningful music-making.

30. Assessment

Assessment should include spelling, construction, identification, performance, transposition, analysis, voice leading, hearing, and application according to objective scope.

31. Mastery

Mastery is multidimensional and context-specific. A learner may perform a shape fluently while lacking note knowledge, hearing, or transfer; records must preserve that profile.

32. Review

Review should vary keys, roots, positions, voicings, tempos, progressions, and genres. Repertoire use may count only when the target relationship is actually observable.

33. Error Classification

Errors include formula error, spelling error, root confusion, degree confusion, inversion error, voicing substitution, function overgeneralization, timing breakdown, voice-leading discontinuity, and stylistic mismatch.

34. Remediation

Remediation should return to the smallest missing relation: interval formula, note spelling, root identification, guide-tone path, rhythm, or auditory model, then restore context.

35. Adaptive Difficulty

Variables include key, root, collection size, position, voicing density, progression length, tempo, accompaniment complexity, constraint count, and amount of notation or audio support.

36. Creative Evaluation

Open-ended work should be evaluated against objectives, constraints, coherence, intentionality, and revision evidence rather than taste conformity.

37. Accessibility

Equivalent demonstrations may use notation, MIDI, adapted instruments, verbal analysis, or recorded performance when they preserve the target. Physical grip limitations must not be confused with harmonic understanding.

38. Anti-Patterns

Noncompliant patterns include scale boxes without degree content, chord dictionaries without construction, Roman numerals without sound, mode names without tonal centers, improvisation as random note selection, and analysis detached from performance.

39. Constitutional Non-Negotiables

1. Scales **MUST NOT** be reduced to fingerings.
2. Chords **MUST** be distinguished from specific voicings.
3. Harmonic labels **MUST** connect to sound and context.
4. Application **MUST** require intentional musical choice.
5. Broad claims **MUST** include transposition or varied roots.
6. Rhythm and ear integration **MUST** occur where the capability is intended for music-making.
7. Creative work **MUST** allow multiple valid outcomes.
8. Repertoire **MUST** support, not replace, objective-aligned evidence.

40. Conformance Tests

Reviewers must identify the structure, reference frame, audible target, physical realization, temporal condition, application task, evidence limits, review plan, and transfer requirement.

41. Tradeoffs and Resolution Rules

Breadth versus depth: prioritize generative relationships over exhaustive catalogs.

Theory versus immediate music-making: introduce enough structure to support meaningful use, then deepen through spiraling return.

Style specificity versus generalization: teach broad principles and mark stylistic boundaries explicitly.

42. Assumptions Made

This chapter assumes Western tonal and modal materials as major initial content while recognizing that FretGarden may later broaden its musical systems and repertoire.

43. Unresolved Questions

Unresolved matters include genre scope, default key sequences, voicing mastery thresholds, repertoire licensing, creative rubric calibration, and automated harmonic analysis limits.

44. Topics Intentionally Deferred

Advanced jazz harmony, species counterpoint, orchestration, atonal theory, world-music systems, and professional composition portfolios are deferred.

45. Required Future Cross-References

Instructional voice, accessibility, quality assurance, analytics, and automated-feedback chapters must preserve integrated musical evidence and creative openness.

46. Chapter Conformance Statement

A compliant system develops scales, chords, and harmony as heard, understood, playable, transferable, and creatively usable relationships rather than static diagrams or labels.